

Fast Topology & Smart Compression for AI

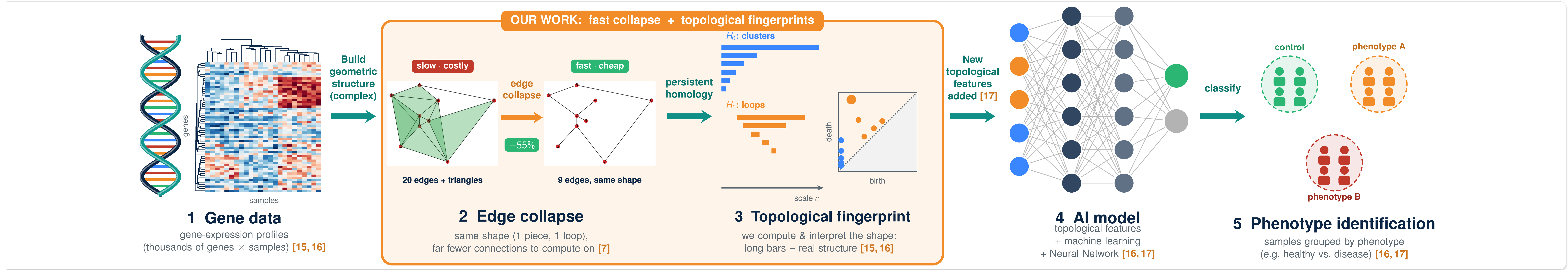
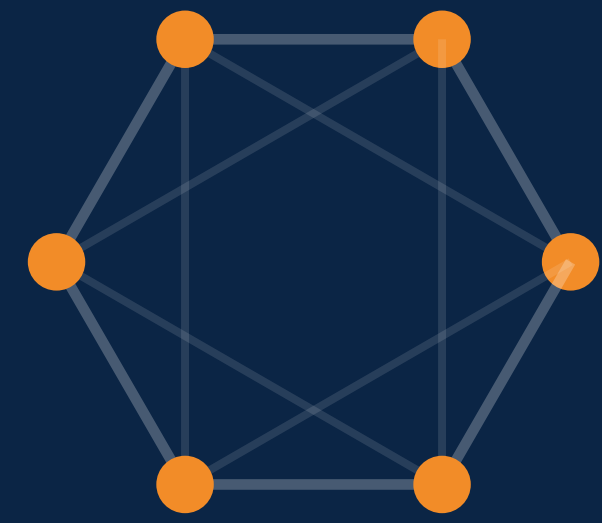
Faster collapse algorithms and dimensionality reduction make data analysis scale

Szybka topologia i inteligentna kompresja danych dla AI

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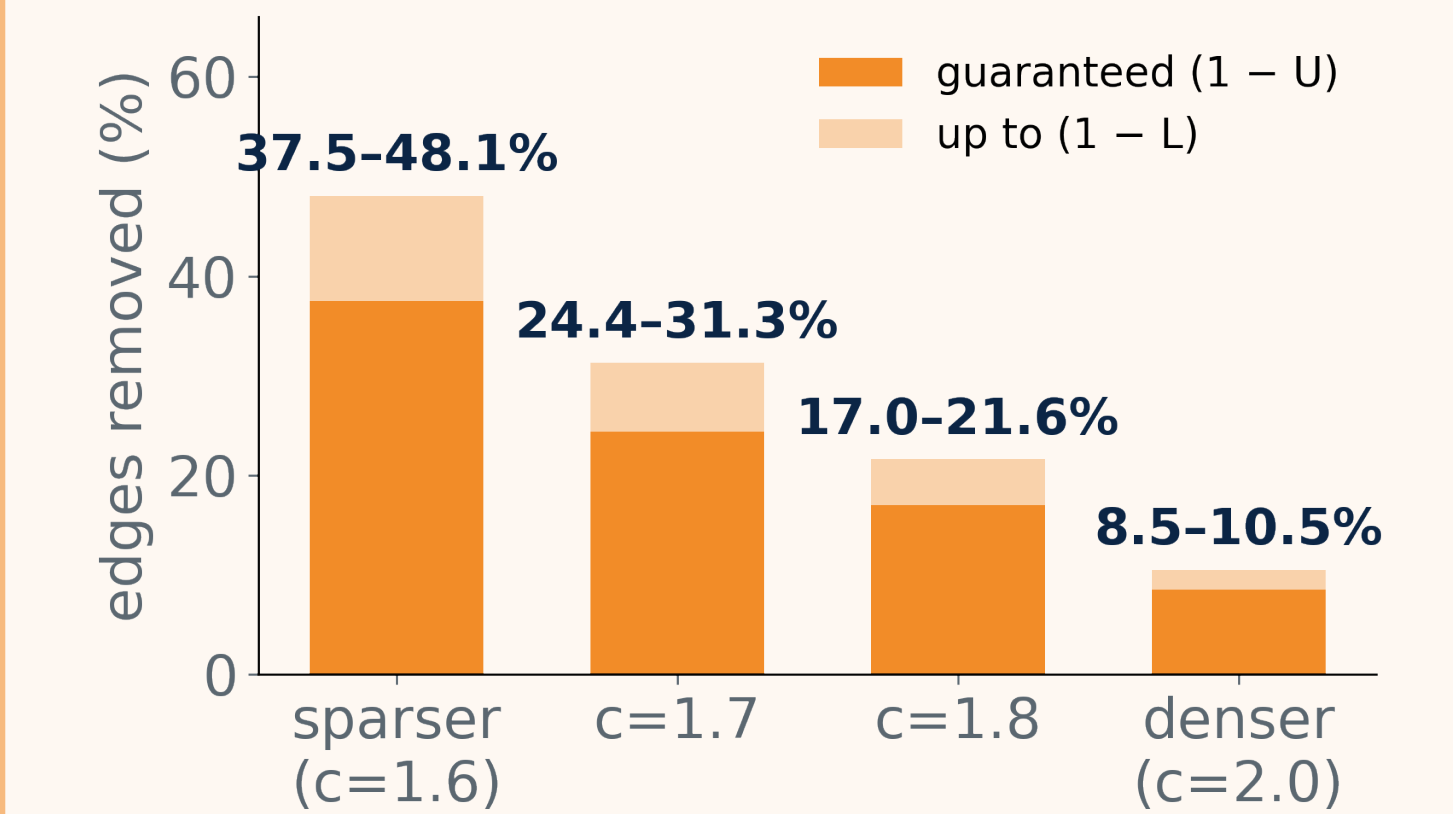
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Our research: faster topology, smarter compression

OUR WORK

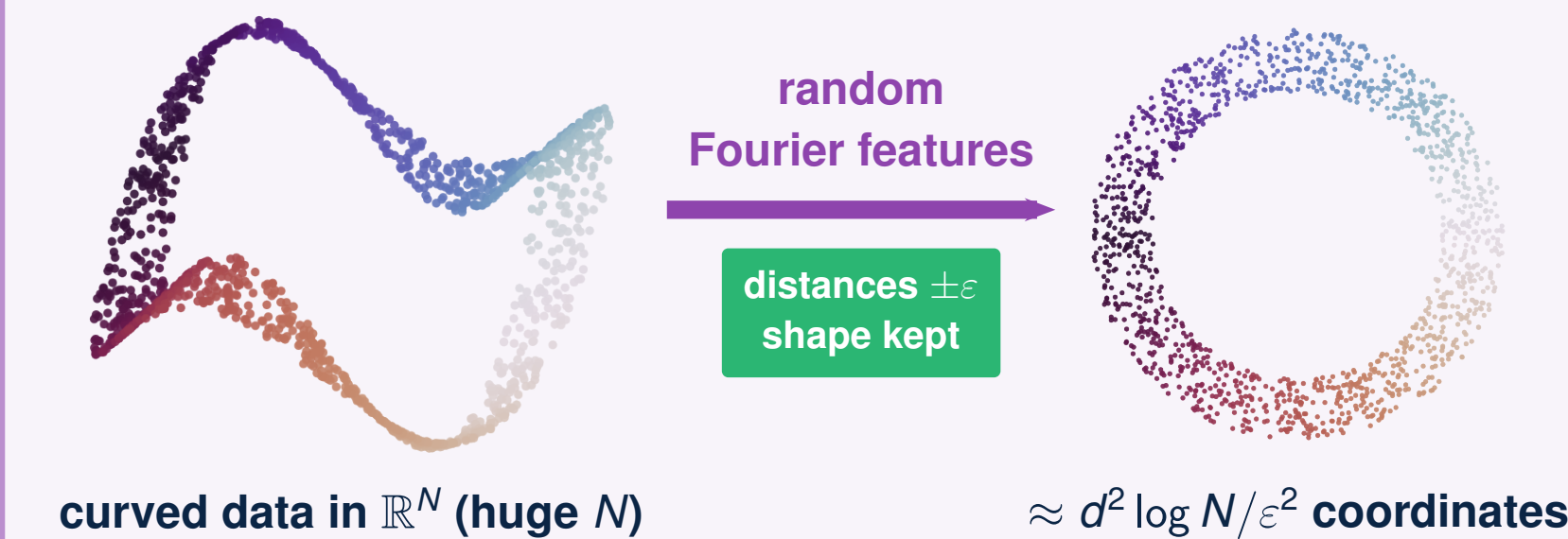
A Edge collapse – proven savings (SoCG 2024, with J.-D. Boissonnat, K. Dutta, S. Pritam [7])



Our theorem [7] for random networks $X(n, c/\sqrt{n})$, a.a.s. as $n \rightarrow \infty$:
 $L \mathbb{E}[e(X)] \lesssim e(\text{core}) \lesssim U \mathbb{E}[e(X)]$, with $U = 1 - \frac{c}{e} (1 - (1 - \gamma)^2)$,
 $L = 1 - \gamma - c^2 \gamma (1 - \gamma)^2$, γ = smallest positive root of $x = e^{-c^2(1-x)^2}$. Bars: 1-U to 1-L.

- ▶ **Smaller input, lower bill:** topology software slows sharply as data grows, so every connection removed saves time and memory [6].
- ▶ **Savings you can forecast:** the first mathematical guarantee of how much edge collapse removes – known *before* you pay for compute.
- ▶ **Same answer, ready to use:** Useful topological features are kept exactly; edge collapse ships in the open-source GUDHI library [18].

B Topology-preserving dimensionality reduction for AI (with K. Dutta, arXiv 2026 [13])



- ▶ Kernel methods (kernel PCA, spectral clustering) get **costly** on data with thousands of dimensions.
- ▶ **Our result [13]:** for data on a d -dimensional manifold, about $d^2 \log N / \epsilon^2$ random features keep **every** kernel distance within error ϵ – size grows only **logarithmically** in N – and **persistent homology is preserved**.
- ▶ **Next:** terminal and online embeddings [14] (ongoing).

One engine, many markets

Financial crime. TDA-based anti-money-laundering at HSBC: **>20%** fewer false alarms (vendor-reported) [8].

Stock markets. A topological early-warning indicator rose for **~250 trading days** before the 2000 and 2008 crashes [9].

Drug discovery. Topology + deep learning beat other methods at predicting **drug-protein binding** [10].

Diabetes. **3 patient subtypes** in 11 210 records [1]; **98.7%** accuracy on eye screening [2].

Generative AI. Topology scores **evaluate and guide** generative models [11, 12].

Why it pays off

- ▶ **Proven market:** the Mapper algorithm [5] was patented and commercialised (Ayasdi, now SymphonyAI) [4] and used in banking [8] and hospitals [3].
- ▶ **Explainable AI:** features describe real structure – vital where decisions must be justified.
- ▶ **Faster & cheaper:** our collapse results cut the cost of scaling TDA to big data.

Let's build it together!

We seek **data partners, pilots and co-funding** in finance, pharma, healthcare, media and the public sector.

Szukamy partnerów wdrożeniowych – porozmawiajmy!

